Tyler Chisholm

Real-Time 3D Developer · UE5 Simulation

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UE5 C++ developer with a real-time rendering and VR background, graduating December 2026. I build interactive 3D systems end to end (physics, streaming worlds, VR input, tools) and measure performance with real instrumentation. U.S. citizen, eligible for a security clearance.

Education

M.S. Interactive Entertainment, Florida Interactive Entertainment Academy (FIEA), University of Central Florida. Dec 2026
12 games built across 4 semesters in UE5, Unity and VR.

B.B.A. Finance & Entrepreneurship, Stetson University, DeLand, FL May 2024

Skills

Real-Time	UE5 C++ and Blueprint, World Partition streaming, Chaos physics, OpenXR/VR, Live Link, GPU profiling, performance budgeting
Engines	Unreal Engine 5, Unity, Godot (engine fork), custom Vulkan renderer
Languages	C++, C#, Rust, HLSL, Python
Tools	Perforce, Git, Jira, Visual Studio, Rider, Nsight Graphics, FFmpeg

Experience

Design Lead — Everent (Studio Kismet, FIEA capstone)

UE5.7 · C++ · Blueprint · Chaos · Perforce 2025 – Present

Momentum-based hoverboard game across an open world of floating sky islands in UE5.7 (Nanite, Virtual Shadow Maps, World Partition), coming to Steam.

- Designed and built speed-gated destruction in C++ on Chaos Geometry Collections: a predictive sweep measures impact speed into the wall, drops blockers a frame early to keep momentum, and fires strain, directional and radial velocity fields. It doubles as secret gating and skill expression.
- Built the Destructible Wall Browser editor tool: lists every destructible in the level, exposes its tuning and fires a test projectile at a set point and speed, so designers tune without entering Play mode.
- Implemented a C++ corrupt-surface physical material (world-space gravity, angle-scaled repulsion, boost suppression) as the constraint that keeps puzzles solvable in a game where the player can ride any surface.
- Hardened destructibles for World Partition and Data Layer streaming (delegate double-bind fix, overlap resolution at BeginPlay, streaming-safe destroy guards).
- Built a local playtest analytics pipeline (FFmpeg, faster-whisper, DeepFace, local Llama 3.1) that turns session recordings into transcripts, emotion timelines and coded feedback in the studio's Excel format.

Project Lead — Studio Microdose (FIEA Ventures)

Multiplayer co-op · indie release target 2026 – Present

- Leading a 10-person team on a co-op game set in the Florida Everglades where players catch invasive animals by hand; hands-on in the build while running scope and leads (art, design, tech, community).

Selected Projects

Spatial Engine — ray-traced engine for non-Euclidean spaces

Rust · Vulkan RT · HLSL · OpenXR 2026 – Present

Personal engine, ~155k lines and 680+ commits. I own the architecture, direction and verification; implementation is AI-assisted.

- Hardware ray-traced Vulkan renderer where spaces with different geometry (hyperbolic, spherical, toroidal) connect through portals, with lighting and shadows traced across them.
- Runs the full path tracer in VR at 90 Hz on a Quest over Link on modest RTX hardware, across ~900 km of streamed terrain with volumetric clouds.
- Found and fixed a shader performance regression from GPU timestamp traces by replacing a 64-step sphere tracer with an analytic quadratic solve.

Asymmetric VR rally game

Unity · Quest 3 PCVR · Logitech wheel 2026

- Local co-op driving game pairing a VR driver with a screen-side co-driver; hardware-agnostic input layer (Strategy pattern) for wheel, pedals and controllers, plus seated calibration.

Earlier Experience

Quality Assurance Agent — Stormbreaker Studios, Orlando, FL

Jan – Jul 2020

Tested 8 iOS client apps; wrote 85 defect tickets with repro steps and fix recommendations.